COSMO® Cube — Manufacturing Guide v1.0

Document: COSMO-MANUFACTURING-v1.md
Target: 100-unit production run
Manufacturer: PCBWay (primary)
Date: 2026-06-12

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1. Manufacturing Overview

The COSMO® Cube is produced across three parallel manufacturing streams:

```
Stream A: PCBWay Electronics
% % %  PCB fabrication (4-layer, 35×35mm)
% % %  SMT component placement (turnkey)
% % %  Reflow soldering (lead-free SAC305)
% % %  Post-reflow AOI inspection

Stream B: PCBWay CNC Machining
% % %  316L SS back plate (mirror polished)
% % %  316L SS front bezel (satin + cutouts)
% % %  CNC tolerances ±0.05mm
% % %  Laser engraving

Stream C: Custom Components
% % %  LiPo battery (Grepow, 35×35×2.5mm)
% % %  OLED display modules
% % %  Camera modules (Himax HM01B0)
% % %  Wireless charging coils

Final Assembly (USA or designated facility):
% % %  PCB !’ coil !’ battery !’ enclosure stacking
% % %  Firmware flashing + provisioning
% % %  QA testing (every unit)
% % %  Packaging
```

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2. PCBWay Order — Step by Step

2.1 PCB Fabrication Order

URL: https://www.pcbway.com/orderonline.aspx

Files to upload:

- Gerber files (ZIP): COSMO-PCB-v1-gerbers.zip
  - Copper layers: F.Cu, B.Cu, In1.Cu, In2.Cu
  - Drill files: drill.drl, drill-NPTH.drl
  - Paste layers: F.Paste, B.Paste
  - Silk: F.Silkscreen
  - Mask: F.Mask, B.Mask
  - Outline: Edge.Cuts

PCBWay Order Parameters:

```
Layers:            4
Board Dimensions: 35 × 35 mm
Board Thickness:  0.8 mm
Quantity:         110
Surface Finish:   ENIG
Copper Weight:    1 oz (outer), 0.5 oz (inner)
Min Hole Size:    0.2 mm (laser via)
Min Track/Space:  4/4 mil
Solder Mask:      Black
Silkscreen:       White (top only)
Via Filling:      Filled + capped (IPC-4761 Type VII)
IPC Class:        Class 2
RoHS:             Yes
```

Estimated price (110 boards, 4-layer): $280–$380

2.2 SMT Assembly Order (Turnkey)

URL: https://www.pcbway.com/assembly/

Files to upload:
- BOM file: COSMO-BOM-Assembly.csv (see format below)
- Pick-and-Place file: COSMO-CPL-v1.csv
- Schematic PDF: COSMO-Schematic-v1.pdf

BOM CSV Format for PCBWay:

```
Comment,Designator,Footprint,Quantity,LCSC/Mouser MPN,Supplier,Notes
ESP32-S3-MINI-1U-N8,U1,ESP32S3-MINI-1U,1,713-ESP32-S3-MINI-1U,Mouser,Provide socket
SSD1327 OLED,DISP1,FPC-24P-0.5mm,1,SEE BOM,BuyDisplay,DNP - hand install
BME280,U2,BME280-LGA,1,828-BME280,Mouser,Near mesh cutout
ICM-42688-P,U3,QFN-14-2.5x2.5,1,ICM-42688-P,Mouser,SPI interface
APDS-9960,U4,LCC-6,1,630-APDS-9960,Mouser,Behind display
BQ51013B,U5,VQFN-20,1,595-BQ51013BRHLR,Mouser,
BQ25892,U6,WQFN-24,1,BQ25892RTWR,Mouser,
AP2112K-1.8V,U7,SOT-23-5,1,AP2112K-1.8TRG1,DigiKey,
W25Q64JV,U8,SOP-8,1,W25Q64JVSSIQ,DigiKey,Optional
RGB LED,LED1,1206,1,APTR3216ZGCK,Mouser,
Tactile Button,SW1,SMD-6x6mm,1,EVQ-Q2C03W,Mouser,
100nF cap,C1-C20,0402,20,GCM155R71C104KA55D,Mouser,Decoupling
10µF cap,C21-C25,0603,5,GRM188R61A106KE69D,Mouser,Bulk
10k:’ &W2Å# Õ#  Ã C "Ã  Å$3 C $e"Ó s  ´ÂÄÖ÷W6W"Å VÆÂ×W öF÷và
```

Assembly Parameters:

```
Solder paste: SAC305 (lead-free)
Reflow:       Standard lead-free profile (peak 260°C)
Testing:      100% AOI
X-ray:        On U5 (BQ51013B VQFN), U6 (BQ25892 WQFN)
DNP list:     DISP1, J_CAM (camera connector), J_BAT (battery connector)
```

Estimated assembly cost (110 units): $900–$1,500

2.3 CNC Machining Order

URL: https://www.pcbway.com/rapid-prototyping/manufacture/?type=cnc

Files to upload:
- Part A (Back plate): COSMO-Back-Plate.step + COSMO-Back-Plate.dxf
- Part B (Front bezel): COSMO-Front-Bezel.step + COSMO-Front-Bezel.dxf
- Technical drawing: COSMO-Enclosure-Drawing-v1.pdf

Part A — Back Plate Specifications:

```
Material:     316L Stainless Steel
Dimensions:   40.0 × 40.0 × 0.8 mm (overall)
Features:
  - Flat plate, no through-holes (wireless charging only)
  - 4× M1.0 threaded blind holes, 1.5mm deep, at corners (3mm from each edge)
  - Stepped rim: 0.3mm × 0.3mm rebate around perimeter for gasket
Surface:
  - Face finish: Mirror polished #8 (Ra <0.025µm)
  - Rim finish: Bead blasted satin
Engraving:
  - "LOT®" center, 8mm wide, 0.2mm deep laser etch
  - "COSMO® CIA" lower-right, 4mm wide, 0.15mm deep
  - Serial number lower-left, 3mm wide (variable per unit)
  - Fill all engravings with black epoxy
Qty: 110
```

Part B — Front Bezel Specifications:

```
Material:     316L Stainless Steel
Overall:      40.0 × 40.0 × 4.2 mm (assembled height)
Wall:         0.5mm minimum thickness
Features:
  - Rectangular frame with 5mm rim
  - Central display aperture: 29 × 29mm (glass/OLED seats inside)
  - 4× M1.0 countersunk through-holes at corners
  - Camera aperture: 5mm diameter circle, bottom-left (8mm from left, 5mm from bottom)
  - Button aperture: 8mm diameter circle, bottom-center, 4mm depth recess
  - LED aperture: 2.5mm diameter circle, bottom-right
  - Weather mesh: 5×5mm area, 0.5mm hole grid, right-center
  - EPDM gasket channel: 0.5 × 0.5mm square groove, inner perimeter
  - PCB standoffs: 4× M1.0 × 1.5mm blind post, interior, 5mm from corners
Surface:
  - Exterior: Bead blasted satin (Ra 0.4–1.6µm)
  - Interior: As-machined (no finish)
Qty: 110
```

CNC Order Notes for PCBWay:

> *"We require 316L (not 304) stainless steel for corrosion resistance.*
> *Back plates require mirror polishing to #8 finish — Ra <0.025µm.*
> *Front bezels require bead blast satin finish.*
> *All laser engraving to be filled with black epoxy (standard jewelry technique).*
> *Serial numbers are sequential: CQ-001-26 through CQ-110-26.*
> *Please provide per-unit serial engraving as a variable field.*
> *Requested dimensional tolerance: ±0.05mm.*
> *Request DFM review before production start."*

Estimated CNC cost (110 sets): $4,000–$6,000

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3. Quality Control

3.1 PCB QA Checkpoints

Test	Method	Pass Criteria
Visual inspection	AOI (automated)	No bridges, missing parts, tombstones
Solder joint quality	X-ray (BQ chips)	Full pad coverage, no voids >25%
Power rail voltage	Bench measurement	3.3V ±2%, 1.8V ±2%
WiFi connectivity	ESP-IDF test firmware	-70dBm or better at 1m
Display function	Firmware test screen	All 16 gray levels displayed
Button function	Bench press + scope	Clean falling edge, < 5ms bounce
Sensor readings	Firmware sensor test	Temp ±2°C of reference, humidity ±5%
Wireless charging	Qi pad @ 5W	Full charge current within 30s of placement

3.2 Enclosure QA

Test	Method	Pass Criteria
Dimension check	Caliper measurement	±0.1mm on all critical dims
Mirror finish	Visual + Ra measurement	No scratches, Ra <0.05µm
Engraving depth	Optical profilometer	0.15–0.25mm depth
Assembly fit	Hand assembly test	No gap > 0.1mm when assembled
IP54 splash	5-min water spray test	No water ingress

3.3 System QA (Every Unit)

```
COSMO® Cube QA Checklist — Unit: CQ-___-26

%¡ 1. Flash firmware v1.0.0-001
%¡ 2. Run provisioning tool — assign serial + API key
%¡ 3. Boot — verify boot screen displays
%¡ 4. WiFi connection — verify connects within 30s
%¡ 5. API poll — verify notification received from LOT test server
%¡ 6. Notification display — verify text renders correctly on OLED
%¡ 7. Copy button — verify LED pulse + Log entry on lot-systems.com
%¡ 8. BME280 — verify temp/humidity/pressure within calibrated range
%¡ 9. APDS-9960 — verify ambient light value, proximity response
%¡ 10. ICM-42688 — verify accelerometer reads "H 9.81 Z-axis
%¡ 11. Wireless charging — verify charge current, LED indicator
%¡ 12. Deep sleep — verify wakeup from button press
%¡ 13. Enclosure assembly — secure M1.0 screws (torque: 0.3 N·m)
%¡ 14. Final visual — no scratches on mirror face, clean front

QA Technician: ___________  Date: ___________  Pass / Fail
```

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4. Production Timeline

4.1 Gantt Chart (Target: 10-Week Build)

```
Week | Activity
% % % % % %<% % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %
  1  %  PCB design finalized !’ Gerbers generated
     %  Order: Battery (Grepow) — 8-week lead
     %  Order: SS enclosure (PCBWay CNC) — 3-week lead
     %
  2  %  Order: All SMD components (Mouser/DigiKey)
     %  Order: OLED displays (BuyDisplay) — 2-week lead
     %  Order: Camera modules (ArduCam)
     %  Order: Qi coils + ferrite (Alibaba)
     %
  3  %  PCB + SMT order placed (PCBWay) — 2-week lead
     %  Firmware development begins (ESP-IDF)
     %
  4  %  Components arrive (SMD, sensors)
     %  Firmware: WiFi + API integration
     %
  5  %  PCB boards arrive (PCBWay)
     %  Prototype assembly (10 units)
     %  Firmware: Display driver, button handler
     %
  6  %  Prototype QA — identify issues
     %  CNC enclosures arrive (PCBWay CNC)
     %  Firmware: Sensor drivers, power management
     %
  7  %  Firmware: OTA, security, factory provisioning
     %  LOT backend: hardware API endpoints
     %
  8  %  Battery arrives (Grepow)
     %  Full production assembly begins (100 units)
     %  LOT Log tab: hardware entry display
     %
  9  %  QA testing — all 100 units
     %  Failed units: rework or replace
     %  Packaging
     %
 10  %  100-unit build complete
     %  Documentation finalized (PDF export)
     %  Ship to Vadim / warehouse
```

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5. Engineering Files Required

5.1 For PCBWay PCB Order

```
cosmo-pcb-v1/
% % %  gerbers/
%    % % %  COSMO-F_Cu.gbr          # Front copper
%    % % %  COSMO-B_Cu.gbr          # Back copper
%    % % %  COSMO-In1_Cu.gbr        # Inner layer 1 (GND plane)
%    % % %  COSMO-In2_Cu.gbr        # Inner layer 2 (Power plane)
%    % % %  COSMO-F_Mask.gbr
%    % % %  COSMO-B_Mask.gbr
%    % % %  COSMO-F_Paste.gbr
%    % % %  COSMO-F_Silkscreen.gbr
%    % % %  COSMO-Edge_Cuts.gbr
%    % % %  COSMO-drill.drl
%    % % %  COSMO-drill-NPTH.drl
% % %  COSMO-BOM-Assembly.csv       # BOM for turnkey SMT
% % %  COSMO-CPL-v1.csv             # Pick and place coordinates
% % %  COSMO-Schematic-v1.pdf
```

5.2 For PCBWay CNC Order

```
cosmo-enclosure-v1/
% % %  3D-Models/
%    % % %  COSMO-Back-Plate.step
%    % % %  COSMO-Front-Bezel.step
% % %  2D-Drawings/
%    % % %  COSMO-Back-Plate.dxf
%    % % %  COSMO-Front-Bezel.dxf
%    % % %  COSMO-Enclosure-Drawing-v1.pdf
% % %  Engraving/
    % % %  COSMO-Engraving-Layout.ai  # Adobe Illustrator for laser paths
```

Design tool recommendation: KiCad 8.0 (PCB, free, open-source), FreeCAD or Fusion 360 (enclosure CAD).

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6. Cost Summary

Item	Unit Cost	100 Units	110 Ordered
PCB fabrication	$3.50	$350	$385
SMT assembly	$12.00	$1,200	$1,320
ESP32-S3-MINI-1U	$3.80	$380	$418
OLED display	$5.50	$550	$605
Camera (HM01B0)	$4.20	$420	$462
BME280	$2.80	$280	$308
ICM-42688-P	$3.50	$350	$385
APDS-9960	$2.10	$210	$231
Qi Rx IC (BQ51013B)	$2.80	$280	$308
Qi Rx coil + ferrite	$1.50	$150	$165
Power IC (BQ25892)	$2.60	$260	$286
Battery (custom LiPo)	$6.50	$650	$715
Passives + misc SMD	$3.00	$300	$330
SS Enclosure (2 parts)	$40.00	$4,000	$4,400
Qi Tx pad (charger)	$9.00	$900	—
Packaging	$4.00	$400	—

Subtotal		$10,730	
Contingency 15%		$1,610	
Total		$12,340	
Per-unit cost	$110		

Target retail price: $349 (2.9× markup, reflects premium SS finish + LOT platform value)

Document v1.0 — COSMO® CIA — LOT Systems, Inc.
Inventor: Vadim Marmeladov — 2026-06-12